

Homework 03: Sample solution

QMIR

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1 Code cells in Quarto

Section 1 of the starter is a demonstration and asks for nothing. A submission should leave it as it is and render it without errors.

2 Objects

```
favourite_number <- 7  
home_town <- "Mannheim"  
likes_coffee <- TRUE  
  
favourite_number
```

```
[1] 7
```

```
home_town
```

```
[1] "Mannheim"
```

```
likes_coffee
```

```
[1] TRUE
```

```
favourite_number <- "seven"  
favourite_number
```

```
[1] "seven"
```

`favourite_number` was silently overwritten: it now holds the text `"seven"` and is of class character, and R gave no warning that the old number was gone.

3 Functions

```
values <- c(3, 8, 12, 5, 9)  
  
mean(values)
```

```
[1] 7.4
```

```
min(values)
```

```
[1] 3
```

```
max(values)
```

```
[1] 12
```

```
round(mean(values), digits = 2)
```

```
[1] 7.4
```

```
km_to_miles <- function(km) {  
  km * 0.621371  
}
```

```
km_to_miles(10)
```

```
[1] 6.21371
```

```
km_to_miles(c(5, 10, 42.195))
```

```
[1] 3.106855 6.213710 26.218749
```

The function works on a vector without any change, because multiplication in R works element by element.

```
percent <- function(x, digits = 1) {  
  round(x * 100, digits = digits)  
}
```

```
percent(0.12345)
```

```
[1] 12.3
```

```
percent(0.12345, digits = 3)
```

```
[1] 12.345
```

4 Reading errors

```
countrye <- c("Germany", "France")  
country_count <- length(countryes)
```

Error:
! object 'countryes' not found

```
Sum(c(1, 2, 3))
```

Error in `Sum()`:
! could not find function "Sum"

```
"10" * 2
```

Error in `"10" \* 2`:
! non-numeric argument to binary operator

1. object 'countryes' not found: the error happens inside `length()`, because the object was created as `countrye` (a typo) and `countryes` does not exist.
2. could not find function "Sum": R is case-sensitive, and the function is called `sum()` in lowercase.
3. non-numeric argument to binary operator: "10" is in quotes, so it is text, and R cannot multiply text by a number.

```
countries <- c("Germany", "France")  
country_count <- length(countries)  
sum(c(1, 2, 3))
```

```
[1] 6
```

```
as.numeric("10") * 2
```

```
[1] 20
```

Writing `10 * 2` directly is an equally good fix for the last line.

5 The pipe

```
distances <- c(4.2, 10, 21.1, 42.2)
round(sum(distances), digits = 0)
```

```
[1] 78
```

The nested call adds up the four distances and rounds the total to a whole number.

```
distances |> sum() |> round(digits = 0)
```

```
[1] 78
```

```
distances |> km_to_miles() |> sum() |> round(digits = 0)
```

```
[1] 48
```

Both versions of the first calculation return the same value. The pipe reads left to right in the order the steps happen, which is why the extension only needs one more step at the front.

6 Data types

Predictions: numeric, integer, character, logical, character, logical, Date.

```
class(42)
```

```
[1] "numeric"
```

```
class(42L)
```

```
[1] "integer"
```

```
class("42")
```

```
[1] "character"
```

```
class(FALSE)
```

```
[1] "logical"
```

```
class("FALSE")
```

```
[1] "character"
```

```
class(10 > 5)
```

```
[1] "logical"
```

```
class(as.Date("2026-09-30"))
```

```
[1] "Date"
```

The two values that usually catch people out are "FALSE", which is character because it is in quotes, and `10 > 5`, which is logical because a comparison returns TRUE or FALSE. Plain 42 is numeric (a double), and only the L suffix makes it an integer.

```
as.numeric("2026") + 1
```

```
[1] 2027
```

```
as.numeric("twenty")
```

Warning: NAs introduced by coercion

```
[1] NA
```

`as.numeric("twenty")` returns NA with a warning. NA is how R writes a missing value: R cannot read the word as a number, so the result is unknown rather than wrong.

```
size <- factor(  
  c("small", "large", "medium", "small"),  
  levels = c("small", "medium", "large")  
)  
  
size
```

```
[1] small large medium small  
Levels: small medium large
```

```
levels(size)
```

```
[1] "small" "medium" "large"
```

7 Data structures

7.1 Vectors and matrices

```
cities <- c("Mannheim", "Heidelberg", "Karlsruhe", "Stuttgart")  
length(cities)
```

```
[1] 4
```

```
population_thousands <- c(315, 160, 308, 633)  
population_thousands * 1000
```

```
[1] 315000 160000 308000 633000
```

```
c(3, "four", 5)
```

```
[1] "3" "four" "5"
```

```
class(c(3, "four", 5))
```

```
[1] "character"
```

```
matrix(1:6, nrow = 2, ncol = 3)
```

```
      [,1] [,2] [,3]  
[1,]     1     3     5  
[2,]     2     4     6
```

A vector holds one type only, so mixing a number with text turns everything into character: "3", "four" and "5".

7.2 A small dataset

```
europe <- data.frame(  
  country = c(  
    "Germany", "France", "Italy", "Spain", "Poland",  
    "Sweden", "Norway", "Switzerland", "Turkey"  
  ),  
  population = c(83.4, 68.4, 59.0, 48.6, 36.6, 10.6, 5.6, 9.0, 85.4),  
  eu_member = c(TRUE, TRUE, TRUE, TRUE, TRUE, TRUE, FALSE, FALSE, FALSE),  
  eu_joined = c(1958, 1958, 1958, 1986, 2004, 1995, NA, NA, NA),  
  region = c(  
    "Western Europe", "Western Europe", "Southern Europe", "Southern Europe",  
    "Eastern Europe", "Northern Europe", "Northern Europe", "Western Europe",  
    "Western Asia"  
  )  
)
```

```
str(europe)
```

```
'data.frame':  9 obs. of  5 variables:  
 $ country   : chr  "Germany" "France" "Italy" "Spain" ...  
 $ population: num   83.4 68.4 59 48.6 36.6 10.6 5.6 9 85.4  
 $ eu_member : logi  TRUE TRUE TRUE TRUE TRUE TRUE ...  
 $ eu_joined : num   1958 1958 1958 1986 2004 ...  
 $ region    : chr   "Western Europe" "Western Europe" "Southern Europe" "Southern Europe" ..
```


europa has 9 rows and 5 columns. `country` and `region` are character, `population` and `eu_joined` are numeric, and `eu_member` is logical.

NA marks a year that does not exist, because Norway, Switzerland and Turkey never joined. A 0 would be read as the year 0, and it would silently drag every average of `eu_joined` towards it.

7.3 A list

```
me <- list(  
  name = "Alex",  
  semesters = 5,  
  courses = c("QMIR", "International Relations")  
)  
  
me
```

```
$name  
[1] "Alex"
```

```
$semesters  
[1] 5
```

```
$courses  
[1] "QMIR" "International Relations"
```

```
me$courses
```

```
[1] "QMIR" "International Relations"
```

8 Accessing and exploring data

```
europa[3, ]
```

```
  country population eu_member eu_joined      region  
3   Italy         59        TRUE     1958 Southern Europe
```

```
europa$population
```

```
[1] 83.4 68.4 59.0 48.6 36.6 10.6 5.6 9.0 85.4
```

```
europa$country[europa$population > 50]
```

```
[1] "Germany" "France" "Italy" "Turkey"
```

```
sum(europa$eu_member)
```

```
[1] 6
```

```
europa$country[europa$eu_joined > 2000 & !is.na(europa$eu_joined)]
```

```
[1] "Poland"
```

```
summary(europa$population)
```

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
5.60	10.60	48.60	45.18	68.40	85.40

```
mean(europa$eu_joined)
```

```
[1] NA
```

```
mean(europa$eu_joined, na.rm = TRUE)
```

```
[1] 1976.5
```

- Four countries have more than 50 million inhabitants: Germany, France, Italy and Turkey.
- `sum()` of a logical vector counts the **TRUE** values, because **TRUE** counts as 1 and **FALSE** as 0, so 6 of the nine countries are EU members.
- Only Poland joined after 2000. Without `!is.na()` the filter also returns **NA** rows for the three non-members, because R cannot say whether a missing year is after 2000. `europa$country[which(europa$eu_joined > 2000)]` is an equally good answer.
- `mean()` returns **NA** as soon as one value is missing. The help page `?mean` documents `na.rm = TRUE`, which drops the missing values first, and gives an average accession year of about 1977 for the six members.

Reflection

Any honest answer is complete. The reflection is read to improve the course, and it is not assessed.

Session info and execution time

```
sessionInfo()
```

```
R version 4.6.1 (2026-06-24 ucrt)
Platform: x86_64-w64-mingw32/x64
Running under: Windows 11 x64 (build 26200)
```

```
Matrix products: default
  LAPACK version 3.12.1
```

```
locale:
[1] LC_COLLATE=English_United States.utf8
[2] LC_CTYPE=English_United States.utf8
[3] LC_MONETARY=English_United States.utf8
[4] LC_NUMERIC=C
[5] LC_TIME=English_United States.utf8
```

```
time zone: Europe/Berlin
tzcode source: internal
```

```
attached base packages:
[1] stats      graphics  grDevices  utils      datasets  methods    base
```

```
loaded via a namespace (and not attached):
 [1] digest_0.6.39    fastmap_1.2.0    xfun_0.60        glue_1.8.1
 [5] knitr_1.52       htmltools_0.5.9  rmarkdown_2.32   lifecycle_1.0.5
 [9] cli_3.6.6        vctrs_0.7.3      compiler_4.6.1   tools_4.6.1
[13] evaluate_1.0.5   pillar_1.11.1    yaml_2.3.12      otel_0.2.0
[17] rlang_1.3.0      jsonlite_2.0.0
```

```
end_time <- Sys.time()
cat(
  "Total execution time:",
```

```
round(difftime(end_time, start_time, units = "mins"), 1),  
      "minutes\n"  
)
```

Total execution time: 0 minutes